NHS

INTERNAL BRIEFING · FROM FIRST PRINCIPLES

Model ED Optimiser

A digital twin of an NHS emergency department — **test every change in the model before you test it on patients.**

This deck assumes no prior knowledge. We will cover: the problem · every screen of the product · how the engine works (in plain English) · what data it needs · how we know it can be trusted · and a jargon-buster.

AGENDA

What we'll cover in the next 30 minutes

1. **The problem**
   why EDs struggle, and why spreadsheets can't answer the hard questions
2. **The blueprint**
   NHS England's "Model Emergency Department" in one slide
3. **The product tour**
   every screen: control tower, workforce, gaps, set-up, exec, planner, analyst
4. **How the engine works**
   two simple ideas — a patient simulator and a hospital "bathtub" — joined together
5. **Trust & validation**
   how we know the numbers aren't made up — including a real-trust test
6. **Data**
   what it runs on out of the box, and how a trust's own data gets in — no coding
7. **Jargon buster + architecture**
   every term in plain English; one slide of tech for the engineers
8. **Where next**
   roadmap and the ask

1 · THE PROBLEM

Why emergency departments jam — the 60-second version

- Patients arrive in a **daily wave** (quiet at 4am, peak early afternoon) — but many EDs staff a flat rota.
- The sickest must be seen first (**acuity**), so lower-acuity patients queue — some give up and **walk out unseen**.
- The biggest cause of gridlock isn't inside the ED at all: when the **hospital's wards are full**, admitted patients can't move out of the ED. They "board" in cubicles. That's called **exit-block**.
- Blocked cubicles mean ambulances can't hand over → **queues at the door**.

The 4-hour standard

% of patients admitted, transferred or discharged within 4 hours. Constitutional target 95%; interim national target 78%. Most trusts miss it.

Decisions made blind

"Add a ward? Open SDEC? Change the rota?" — today these are argued from spreadsheets and instinct. Each choice costs £100k–£2m.

You can't A/B-test an emergency department on real patients. So we built one you can.

The NHS "Model Emergency Department"

NHS England's blueprint (Feb 2026) for how a high-performing ED should be laid out and run. It organises the department by **how sick you are × how long you've been there**, and names the building blocks a good ED should have:

Front door (first 15 min)

Streaming nurse decides where you go. **RAT** — a senior doctor sees the sickest immediately. Ambulance handover within 15–45 min.

Core ED (0–4 h)

Majors cubicles, resus, minor injuries (**UTC**), a seated "fit-to-sit" area, a children's area (**PAU**) — see, treat, decide.

Admission-avoiders (same day)

SDEC, **EEMAC**, **CDU**, frailty unit, mental-health liaison — treat and send home the patients who would otherwise be admitted to a scarce bed.

Most trusts have **some** of these blocks. The question the product answers: **which missing block is worth building, for this specific department, and what is it worth in performance and pounds?**

What we built: a digital twin, in plain clothes

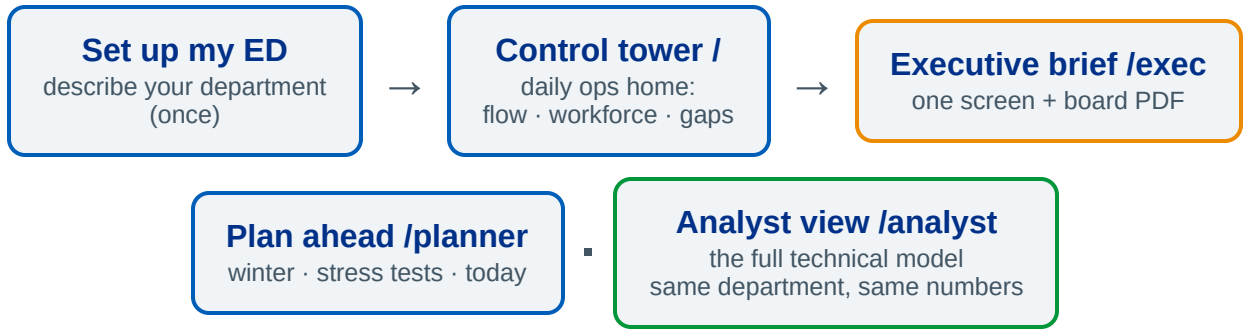
A **digital twin** = a working computer copy of your department. Simulated patients arrive in the same daily wave as yours, queue for the same cubicles and clinicians, and leave through the same routes — so you can change anything and watch what happens, safely.

- **Flow-first & plain-language** — the user sees queues, pinch points, standards, pounds. All simulation jargon is hidden.
- **One saved department** — set it up once; every screen (ops, exec, planner, analyst) models the same ED and agrees with the others.
- **Offline by design** — no cloud calls; runs on locked-down NHS networks.
- **Every assumption visible and editable** — no black box.

Who uses it

Role	They use it for
ED matron / ops manager	today's flow, staffing gaps, the pinch point
Clinical lead	does our layout match the Model ED?
COO / director	risk, standards, investment case, board paper
ICB / winter planner	demand forecasts, stress tests, options
Analyst	the full model, calibration, validation

Four surfaces, one department



Everything reads the **one saved department** — so the number a matron sees on the control tower is the same number the analyst can defend, and the same number in the board PDF.

3 · THE PRODUCT TOUR — SET UP

Describing your ED takes minutes, not a data project

Three no-code routes in — pick whichever is easiest. Anything you don't provide falls back to a sensible, clearly-flagged national default.

ROUTE 1 · FASTEST

Pick your trust

Choose your trust from a dropdown. Published NHS figures (attendances/day, bed numbers, occupancy) pre-fill the model — every value carries its source and date.

ROUTE 2 · FULLEST

Import an Excel workbook

Download the template, fill in tabs for areas, opening hours, rota, costs — optionally arrivals-by-hour and acuity mix — and upload. The whole department in one file.

ROUTE 3 · HANDS-ON

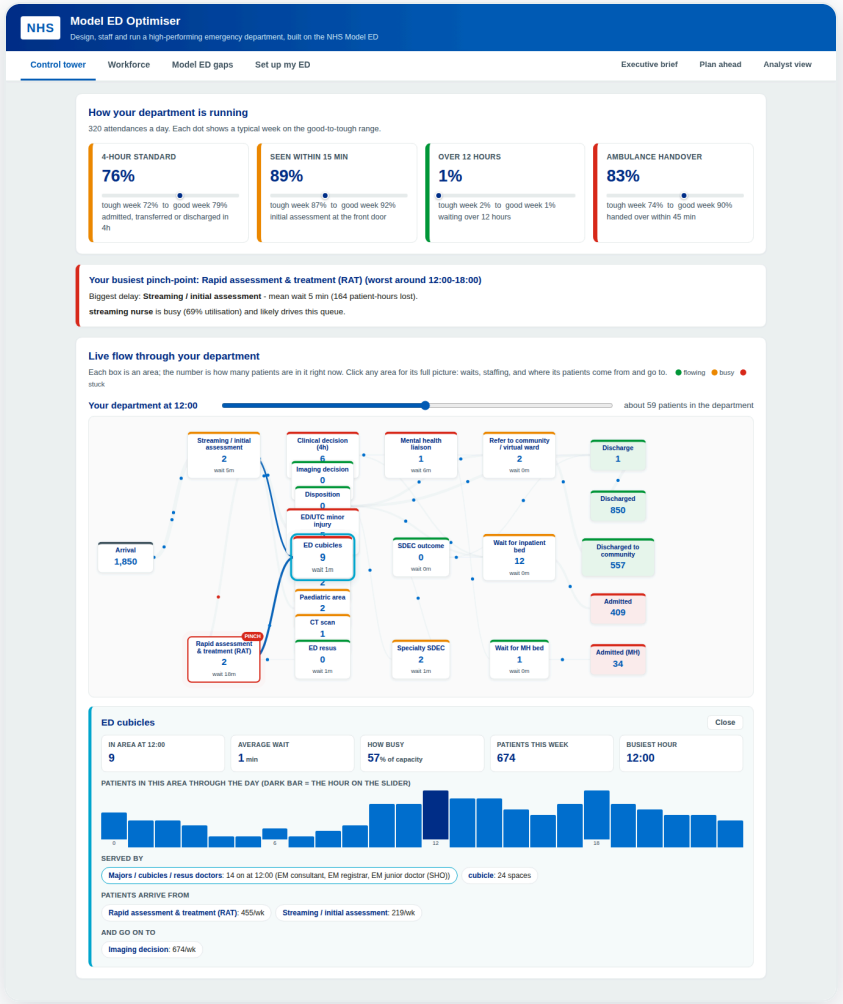
Fill in the form

Toggle the 9 Model ED areas on/off, set each one's opening hours (e.g. SDEC 08:00–20:00), beds, community capacity, and edit the rota shift-by-shift.

Patients re-route safely when an area is switched off or closed for the night — the model never strands anyone, so any configuration you can click is a valid department.

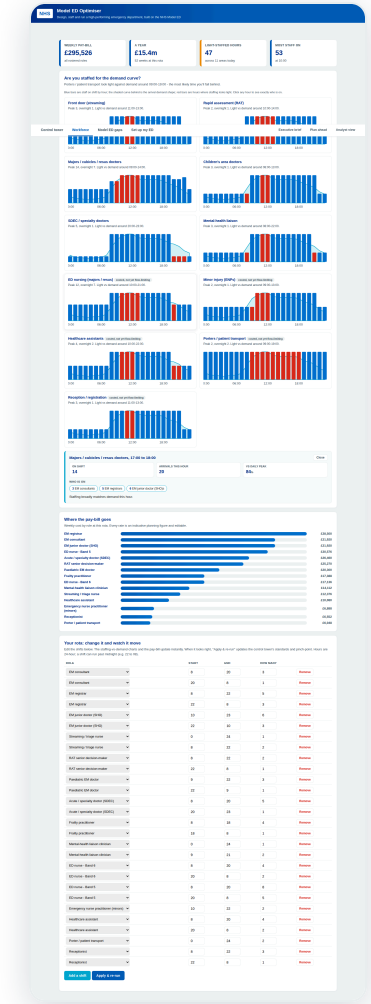
Control tower: where is it stuck, and when?

- **Live flow map** — a box per area, simulated patients moving between them; colours show queueing (green / amber / red).
- **The pinch point** — the stage losing the most patient-hours, flagged, with **when in the day it bites** (e.g. 14:00–18:00).
- **Time-of-day slider** — scrub to any hour; see census and each area's load.
- **Click any box** — where its patients come from, where they go, who's on shift there.
- **Headline standards** — 4-hour, 15-min assessment, 12-hour, handover — each shown as a **good / typical / tough week**, not a single misleading number.



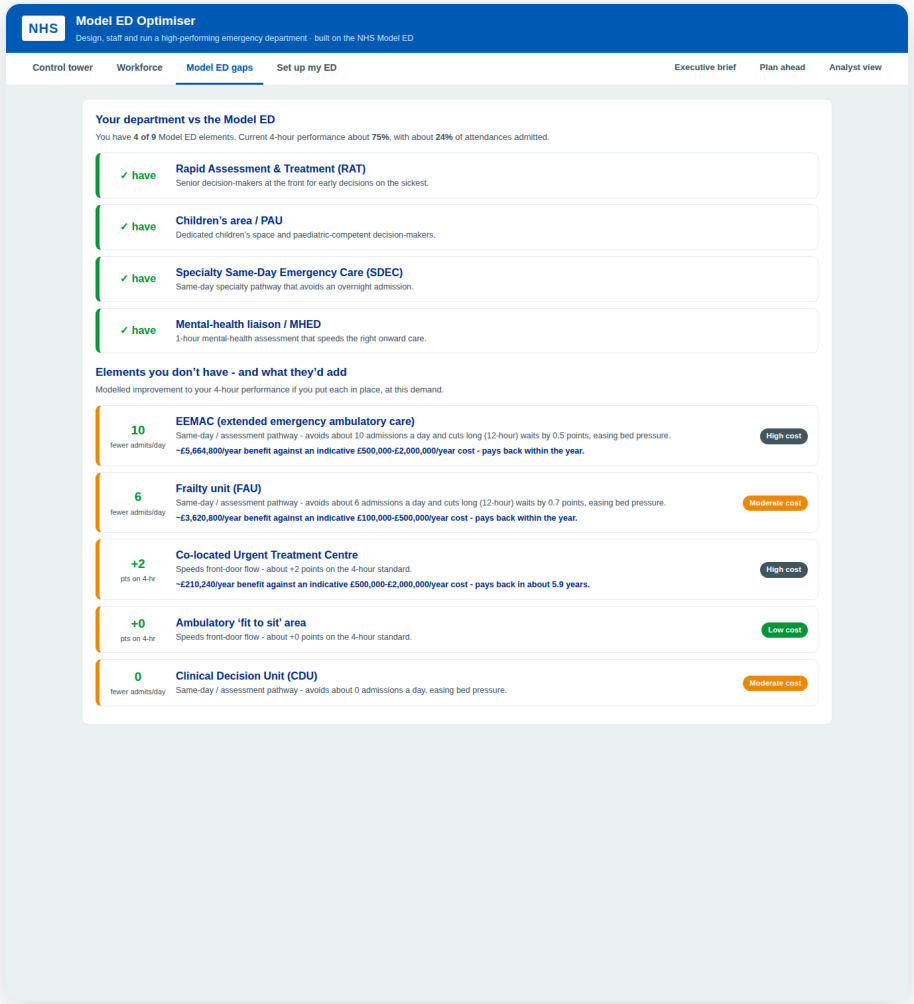
Workforce: are you staffed for the demand curve?

- For every staff group: **people on shift, hour by hour, against the arrival wave**. Under-staffed windows show red.
- The **weekly pay-bill** and where it goes, by role.
- **Live rota editor** — change a shift and the staffing curve + cost update instantly (no simulation needed for the preview).
- **Apply & re-run** — pushes the new rota through the full model: watch the standards and the pinch point move.
- The engine models **staffing that varies by hour** — and a safety rule: cubicles only count as open if there are enough nurses on shift to cover them (1 nurse : 4 cubicles).



Model ED gaps: what's missing, and what it's worth

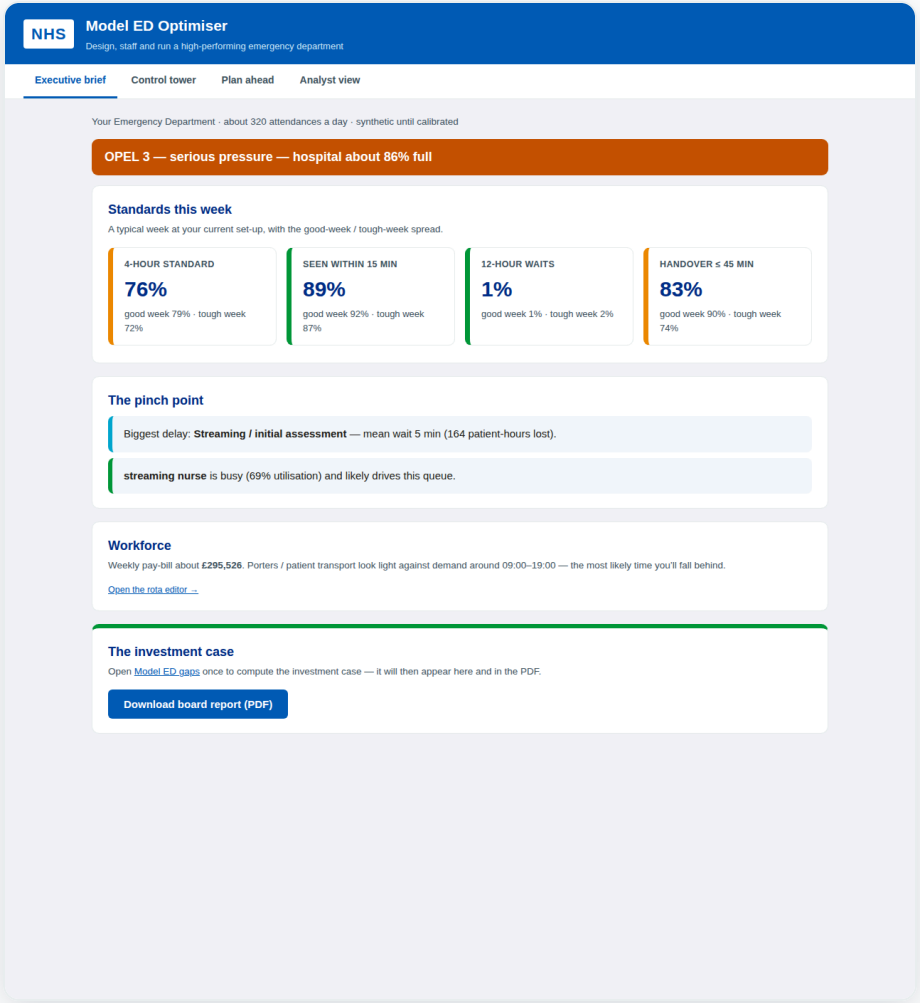
- Compares **your configured areas** against the Model ED blueprint: have / missing.
- For each missing element, the model **simulates your department with it added**: front-door areas judged on the 4-hour standard, same-day areas on the **admissions they avoid**.
- Each gets an indicative cost band (Low / Moderate / High) and a **costed business case**: annual benefit, ROI, payback.
- Every £ comes from a **trust-editable cost book** — and the assumptions are printed on every case. Nothing is presented as a benchmark.



Executive brief: one screen, one click to a board paper

- **OPEL-style risk level** for the configured department.
- The four headline standards as **typical / good / tough** weeks.
- The pinch point and the recommended action, in one sentence.
- The weekly pay-bill and the **single biggest investment case**.
- **One-click board-ready PDF** — the decision report: standards, diagnosis, uncertainty ranges, and the costed investment case with its assumptions.

Written for a COO with 90 seconds — everything else is one click deeper.



Plan ahead: winter, shocks, and this afternoon

Winter plan

Set the winter uplift (e.g. +12% demand). The model forecasts your position on the standards and ranks **costed options** — more beds, community capacity, UTC, staffing — by what they buy back.

Stress tests

Named scenarios: **bad flu season · norovirus ward closures · medical strike day · heatwave**. Each shows the hit (e.g. "−9 points on 4-hour") — and an **inverse solver** answers "what staffing would hold the line through it?"

Today (intraday)

Type in arrivals so far this morning + how full the hospital is. It projects the rest of today and suggests which of the **actions you can actually take this afternoon** helps most.

Crucially: plans model **your department exactly as configured** — your areas, your opening hours, your rota — not a generic template. There's also a demand **forecaster**: upload daily history, it learns trend, weekday and seasonal patterns, and always shows its own accuracy on held-back data.

Analyst view: the workings behind the headlines

Same department, full technical depth — this is where an informatics team checks, challenges and calibrates the model. Six tabs:

Live simulation

Full scorecard, bottleneck diagnosis, the flow map (click any block for volumes / waits / utilisation), Sankey & heatmap.

System Dynamics

The hospital around the ED: bed occupancy, crowding and feedback over time — **with a plain-English interpretation** under every chart.

Monte-Carlo

Runs many simulated weeks → **confidence intervals** on every standard. "78%, but plan for 71–85" beats a single number.

Scenarios & sensitivity

What-ifs vs baseline + a tornado chart of which lever matters most — with a written impact read-out.

Validation

Proof every simulated journey is clinically legal, the common pathway routes with **time at each stage and the total**.

Data & calibration

Upload a real attendance extract (ECDS-style CSV) → arrivals, case-mix and admission rates become **your** data.

Under the bonnet: two simple ideas, joined

Idea 1 — simulate every patient (the "DES")

Like a flight simulator for the department: thousands of individual virtual patients arrive, queue, get treated, move on. Great at queues, staffing and "who waits where".

Idea 2 — simulate the hospital around it (the "System Dynamics")

The wards, mental-health beds and community services as filling-and-draining tanks. Great at the big slow pressure that actually causes gridlock.

Why both? Because the worst ED problem — exit-block — is caused outside the ED. A patient-only simulator literally cannot see it. A tanks-only model can't see queues or rotas. Ours couples the two, every simulated hour.

On top of that: **Monte-Carlo** (run many weeks, because no two weeks are alike) and **validation** (prove the simulated journeys follow the real clinical pathway). The next four slides take each in turn.

DES in plain English: a supermarket with triage

Discrete-Event Simulation (DES): create virtual patients one at a time and walk each through the department, minute by minute. Every queue forms for the same reason a supermarket queue does — more people than open tills. Except here:

- **The sickest jump the queue.** Every resource serves by clinical priority (acuity 1 first) — exactly as an ED does.
- **A cubicle is held, not borrowed.** A patient keeps their cubicle across X-ray and review until they leave — meanwhile doctors and nurses are borrowed stage by stage. Both are contended at once.
- **Staffing varies by hour.** The rota is the model: 3 doctors at 4am, 14 at 2pm.
- **People give up.** If the wait exceeds a patient's patience they walk out unseen (LWBS) — long queues create invisible clinical risk, and the model shows it.
- **Ambulances are physical.** A crew occupies a handover bay until the patient is accepted — crowding queues real vehicles.

What a virtual patient looks like

Arrives 14:07, by ambulance · **acuity 2**
→ handover bay 12 min
→ streaming nurse 6 min
→ RAT senior review 15 min
→ **cubicle acquired** · bloods, imaging
→ decision: admit
→ waits for a ward bed… **(this wait depends on the whole hospital — next slide)**
→ transfer · cubicle released · LOS 6h 40m

The pathway itself — which stages exist and how they connect — is a faithful copy of the NHS Model ED design, stored as data and validated.

System Dynamics in plain English: the bathtub

Think of the hospital's ward beds as a **bathtub**: admissions pour in from the top (from the ED *and* from everywhere else), discharges drain out of the bottom. The water level is **bed occupancy**.

- NHS hospitals run at ~92% full. Above that, finding a bed takes longer and longer — the tub is nearly overflowing.
- Our model gives the tub realistic plumbing: a steady background inflow (elective & other admissions) and a drain that speeds up or slows down with how full it is.
- Two more tanks: **mental-health beds** and **community / virtual-ward capacity** — spare community capacity actively **pulls** patients out of hospital, draining the tub faster.

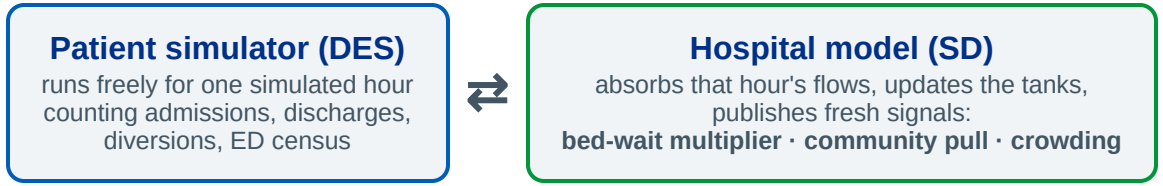
The feedback loop: tub nearly full → ward bed waits stretch → admitted patients board in ED cubicles → cubicles blocked → queues everywhere → ambulances stack at the door.

That chain is exit-block, and in our model it emerges from the physics — nobody typed it in.

270 → 595 min

Admitted patients' time in the ED, feedback off vs on — the model reproduces exit-block from the bed stock alone. Switch feedback off and you get the naïve model back.

How the two models talk: an hourly handshake



EVERY SIMULATED HOUR

1 · Drain

The hour's patient flows are handed from the DES to the hospital model.

2 · Update

The tanks integrate: occupancy moves, the bed-wait multiplier is recomputed and smoothed.

3 · Publish

The new signals are pinned for the next hour; the DES reads them (e.g. bed waits stretch ×2.5) but can't disturb them mid-hour.

For the engineers: this is a **zero-order-hold co-simulation** — hold signals constant between exchanges, use smooth bounded feedback plus first-order smoothing, and the loop is provably stable (we test for oscillation in CI).

Monte-Carlo: because no two weeks are the same

One simulated week is one **roll of the dice** — a lucky week or a horrible one. So we run the same department **many independent weeks** (different random arrivals each time) and report the spread.

- Every standard gets a **95% confidence interval**, not a point estimate.
- The product shows it as **good / typical / tough week** — language any board understands.
- Wide spread is itself a finding: *"plan to the lower bound".*

"4-hour: 78%"
what a one-run tool tells you

"Typical 78% · good 85% · tough 71%"
what we tell you — across 24 independent simulated weeks

The recommendation engine goes further: it tests each candidate change with paired weeks (same luck, with and without the change) and only recommends when the improvement is statistically real. Silence over noise.

5 · TRUST & VALIDATION

How we know it isn't making things up

1 · Clinical conformance

Every simulated patient's journey is checked, step by step, against the intended Model ED pathway graph. A clean run passes **100%**; deliberately injected errors are caught. The pathway can't silently drift from the clinical design.

2 · Emergent physics

Exit-block, walk-outs under starvation, priority for the sickest, cubicles capped by nursing — these behaviours **emerge** from the model and are asserted in CI on every change, not hand-tuned in.

3 · A real trust

We fed in Barnsley's published figures — **290 attendances/day, 91.8% bed occupancy** — and nothing else. The model produced **71.3%** on the 4-hour standard. Barnsley's real reported figure: **67.5%**. Within 4 points, untuned.

Observed performance (like that 67.5%) is never fed in as an input — it is held back and used only to test the model. That separation is what makes the validation honest.

Plus ~38 end-to-end checks run in CI on every code change — the test suite is the contract.

6 · DATA

What it runs on, and how your data gets in

Layer	Out of the box	Replace with your own — no coding
Department shape areas, hours, beds, rota	a realistic mid-size template ED	trust picker (published NHS data) · Excel workbook · the set-up form
Demand & case-mix arrival wave, acuity, cohorts	synthetic, shaped like national patterns	upload an **ECDS-style CSV** → arrivals-by-hour, acuity mix, ambulance share, paediatric/frailty/MH prevalence, admission rates all become yours
Money role rates, value of avoided harm	indicative planning defaults, clearly flagged	the cost-book screen, or the workbook's **Costs** tab
Standards RAG targets (e.g. 4-hour 78%)	current national standards	editable via API as policy moves (78% → 95%)
Model physics how long each stage takes	expert-set estimates, flagged as such	a global or per-stage adjustment (API / workbook **ServiceTimes** tab) — tuned until the model reproduces your observed performance

Honest limits: stage-by-stage treatment times are not recorded in any national dataset, so they remain calibrated assumptions — visible, editable, and validated against your observed outcomes. The pathway *structure* itself stays version-controlled (that's what keeps it clinically conformant).

6 · WHAT YOU WALK AWAY WITH

The outputs, in one slide

A scorecard against the standards

4-hour · 15-min initial assessment · 12-hour waits · ambulance handover ≤45 min · walk-outs (LWBS) · MH liaison ≤1 h · children's 4-hour · EEMAC conversion · bed occupancy — RAG-rated, with good/typical/tough ranges.

The pinch point + what to do

The stage losing the most patient-hours, when it bites, whether a resource is saturated or it's downstream exit-block — and the recommended first move.

A costed investment case

For every missing Model ED element: modelled benefit, cost band, annual value, ROI, payback — assumptions attached.

A board-ready PDF

One click: risk, standards, diagnosis, uncertainty, the investment case. The same numbers every other screen shows.

Plus: winter plans and stress tests on **your** configuration, an intraday "what helps this afternoon", staffing-vs-demand and the pay-bill, and the full analyst evidence trail behind all of it.

7 · JARGON BUSTER

Every term we've used, in one place

4-hour standard — % of patients admitted, transferred or discharged within 4 h of arrival. Constitutional 95%; interim 78%.

Exit-block — full wards stop the ED admitting; patients board in cubicles and everything backs up.

Boarding — an admitted patient stuck in the ED waiting for a ward bed.

RAT — Rapid Assessment & Treatment: a senior doctor at the front door for the sickest arrivals.

SDEC — Same-Day Emergency Care: specialty assessment that sends patients home the same day instead of admitting.

CDU — Clinical Decision Unit: short-stay observation beds avoiding a ward admission.

MH liaison — the mental-health team in the ED; 1-hour response standard.

DES — Discrete-Event Simulation: modelling every patient individually through queues and resources.

Digital twin — a working computer copy of a real system you can safely experiment on.

Confidence interval — the range results genuinely fall in (our good week / tough week).

Acuity — how sick you are, 1 (life-threatening) to 5 (minor). Drives priority.

LWBS — left without being seen: gave up waiting and walked out. Hidden clinical risk.

Handover — transferring an ambulance patient to ED care; standard is ≤45 min, crews are stuck until it happens.

UTC — Urgent Treatment Centre: minor illness/injury stream alongside the ED.

EEMAC — extended emergency ambulatory care (up to 12 h) for patients expected to go home.

Frailty unit (FAU) — early assessment for frail older patients, preventing unnecessary admission.

ECDS — Emergency Care Data Set: the national record of every ED attendance; our calibration format.

System Dynamics (SD) — modelling the hospital as filling/draining stocks (the bathtub) with feedback.

Monte-Carlo — running the model many times with different random draws to measure the spread of outcomes.

Conformance — the automated check that every simulated journey follows the intended clinical pathway.

Architecture on one slide



- **Pathway as data** — `model_ed.json` defines nodes, routes, resources; the engine is generic. Service-time *values* are a data overlay; only the graph structure is code.
- **Department builder** — a saved `Department` (areas, hours, rota, beds) compiles into a runnable pathway; rota folds into per-hour staffing.
- **Ingest** — trust snapshot CSV (by ODS code), Excel workbook importer, ECDS calibration seam.
- **Persistence** — SQLite (or Postgres): versioned departments, calibrations, settings, and a **run cache** keyed by hash(engine version + exact config) — identical questions answer instantly.
- **Decision layer** — metrics/scorecard, Monte-Carlo, scenarios+tornado, benchmark, forecaster, surge solver, CI-gated recommender, cost book, board PDF.
- **Quality** — ~38-check end-to-end suite in GitHub Actions on every push; conformance + physics asserts are part of it.

Stack: Python (SimPy, NumPy, pandas, Flask, SQLAlchemy) · Plotly (vendored) · React+TS shell · zero external calls at runtime.

What it is not (yet) — the honesty slide

- It is a **planning and design tool**, not a live operational system — no real-time feed yet (the intraday "today" mode is the designed hook for an EPR/FHIR feed later).
- Out of the box it runs on **synthetic, clearly-flagged parameters** — treat results as indicative until calibrated and validated against the trust's own data.
- Stage-level treatment times are **assumptions** (no national dataset records them); they are editable and validated against observed outcomes, but they are not measurements.
- **Single department, no login** today; multi-site and auth are roadmap.
- Governance for real-data pilots (DPIA, clinical-safety case) is **not started** — required before any deployment with patient-level data.

> **Our position: every number carries its assumptions, every assumption is editable, and the model must earn trust by reproducing a trust's own observed performance before anyone acts on it.**

Roadmap, and how to try it right now

Next steps

- **More trusts** in the pick-your-trust registry (each is a data row, not code).
- **React ports** of the remaining surfaces (exec brief already ported).
- **Auth + multi-department** tenancy.
- **Live feed**: EPR/FHIR → the "today" view becomes continuous.
- **Optimiser v2**: budget-constrained search over combinations of moves.
- **Governance pack** for real-data pilots.

Try it in two commands

```
pip install -r requirements.txt
python -m uec_hybrid.app.server
# → http://127.0.0.1:5000
```

Suggested demo: pick a trust → watch the control tower re-run on its real figures → break the rota → watch the pinch point move → print the board PDF. Five minutes, no jargon.